S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

Financial Risk NLP Engine

Author - V Aditya Teja

college - Vellore Institute of Technology

2. Problem & Approach



Problem

Wholesale banking and portfolio teams struggle to digest the massive firehose of unstructured news and social media during fast-moving crisis events (e.g., SVB collapse).

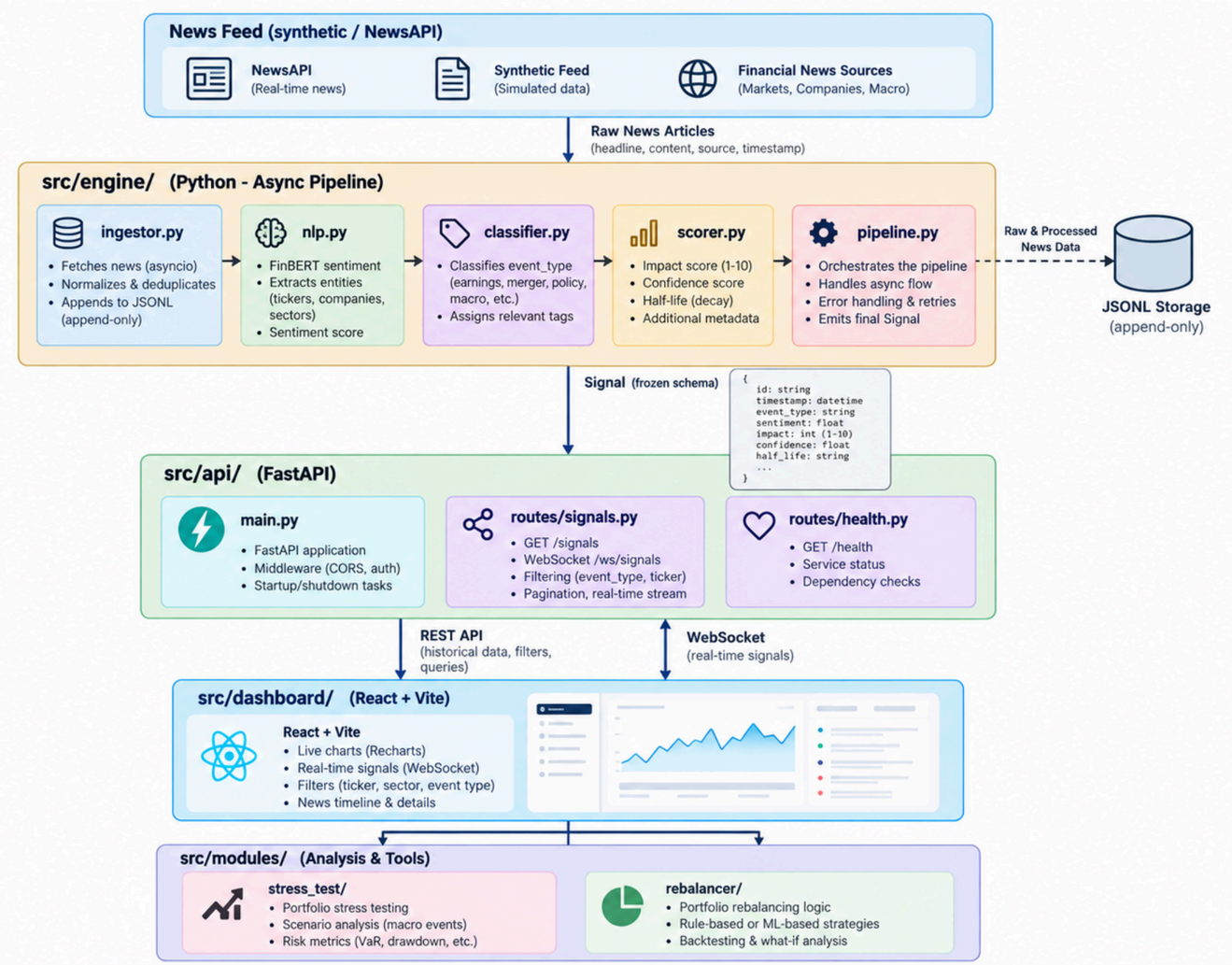


The Approach:

Continuous NLP Pipeline: Ingest streams, perform zero-shot event classification, and calculate FinBERT sentiment.
Explainable Impact Scoring: Translate abstract text into a concrete 1-10 "Impact Score" anchored to historical abnormal asset moves.
Dynamic Rebalancing: Automatically re-weight a portfolio dynamically based on decayed sentiment signals.

3. System Design

- **Ingest**: news and social text through one interface, replayed from history or live
- **Analyze**: FinBERT sentiment, zero-shot event type, calibrated impact (1-10), confidence, evidence
- **Link & Decay**: per-entity state, with each signal fading by its event-type half-life
- **Serve**: one frozen signal schema over REST, WebSocket, and a JSONL file
- **Act**: Module B stress-tests on high-impact events, and Module A rebalances index weights from sentiment



4. Implementation Highlights

1

FinBERT for finance-specific sentiment. It runs locally, so it’s free and fast.

2

Zero-shot event classifier with 8 fixed labels. It needs no training data.

3

Event-study impact calibration: gradient boosting predicts the abnormal return, then buckets it to 1-10.

4

Corroboration weighting: the same claim from several sources raises confidence, and a lone tweet gets low weight.

5

Time-decay by event type: earnings fade in days, credit events persist.

6

Explainable signals: every output carries evidence snippets and sources.

5. Key Results

Metric	Equal Weight	Naive Baseline	NLP Engine
Return	0.99%	0.50%	**0.99%**
Volatility	15.30%	16.00%	**15.30%**
Sharpe	1.59	1.10	**1.59**
Max Drawdown	-3.58%	-4.00%	**-3.58%**

6. Domain Impact

Concrete Use Case: A portfolio manager goes to lunch; while away, geopolitical news breaks. The engine instantly flags the risk and proposes an automated hedge before human intervention.

Who Uses It:

- **Risk Analysts: For live monitoring of counterparty risk.**
- **Quant Traders: As an alpha-generation signal.**



7. Limitations & Next Steps



Limitations (Honest Evaluation):

Zero-Shot Constraints: The DistilBERT NLI model relies heavily on label phrasing and can struggle with ambiguous financial jargon.

FinBERT Blindspots: Struggles with highly contextual sarcasm or extremely novel instruments not in its 10-K training corpus.



Future Work

- **Live APIs:** Connect to Bloomberg or Twitter Enterprise Firehose.
- **Options Pricing:** Integrate Black-Scholes to auto-hedge via put options rather than just re-weighting.
- **LLM Reasoning:** Use GPT-4 to write a textual summary of the daily risk



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The End

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